Aditya Agrawal

Quantitative Research | Software Engineering | AI & Financial Markets
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PROFESSIONAL SUMMARY

First-year Bennett University student building applied projects in systematic trading, filing-grounded AI research, analytics, and interactive software. Strong foundation in probability, statistics, financial mathematics, and Python research workflows, with a focus on making models and tools useful under real-world constraints.

TECHNICAL SKILLS

Programming: Python, C++, TypeScript, JavaScript, Dart
Quantitative Research: Pandas, NumPy, Statsmodels, Kalman filters, backtesting, statistical arbitrage, market-neutral strategies
AI & Data: LLM workflows, prompt engineering, SEC and financial document analysis, financial data APIs
Product Development: React, Next.js, Flutter, Supabase, Firebase, Vercel, Netlify, CSS
Finance: 10-K/10-Q analysis, shareholder letters, execution cost modeling, volatility targeting, portfolio construction

SELECTED PROJECTS

Quantitative Trading Research Framework | Python, Pandas, NumPy, Statsmodels, Kalman Filters | GitHub

- Designed a no-lookahead statistical-arbitrage engine for equity pairs with dynamic Kalman-filter hedge ratios and layered z-score signals.
- Integrated volatility targeting, leverage and ADV constraints, slippage, commissions, borrow costs, and nonlinear market-impact modeling.
- Backtested KO-PEP and MA-V strategies with reported post-cost Sharpe ratios of 1.90 and 1.31 respectively.

AI Financial Research Platform / Deep Asset Research | Python, LLMs, NLP, Financial Data APIs | Project Link

- Built an AI-powered equity-research workspace that produces filing-grounded reports, charts, and investor-ready summaries from 10-Ks, 10-Qs, and shareholder materials.
- Added ticker research, watchlists, current-news views, and saved reports to support repeatable research workflows.

Codeforces Training Dashboard | React, Analytics, Codeforces API, Vercel | Project Link

- Built a coach-style dashboard that analyzes Codeforces history to surface weak topics, rating friction, practice trends, and a tailored next problem set.

Benjamin Franklin: A Life of Curiosity | Next.js, TypeScript, GSAP, React Three Fiber | GitHub

- Created a cinematic, scroll-driven historical experience with archival content, motion-aware interactions, and a reduced-motion alternative.

Interactive Blackjack | React, TypeScript, CSS | Project Link

- Developed a responsive Blackjack game with configurable bankroll, bet, and deck settings; dealer logic; score tracking; and mobile-friendly controls.

Outsmarts | Flutter, Dart, Firebase | Project Link

- Built a cross-platform code-breaking game with guest access, progress tracking, game history, and difficulty-based AI opponents.

RESEARCH INTERESTS

Quantitative finance and systematic investing; statistical arbitrage and market microstructure; AI-driven financial research tooling; insurance, disaster resilience, and risk transfer; development economics.

EDUCATION

First-year student at Bennett University. Self-directed study in advanced probability and statistics, stochastic processes, financial mathematics, development economics, quantitative trading research, artificial intelligence, and software development.